

Compound interest



1 Describe how compound interest is earned.

2 \$9000 is invested for 3 years at a rate of 5% p.a. compounded annually.

a Complete the table:

b Calculate the total interest accumulated over 3 years.

c Calculate the value of the investment at the end of 3 years.

	Interest (\$)	Balance (\$)
After 0 years	0	9000
After 1 year		
After 2 years		
After 3 years		

3 \$3000 is invested at 4% p.a., compounded annually. The table below tracks the growth of the principal over three years.

	Value at start of time period	Value at end of time period	Interest earned
1st year	\$3000	<i>A</i>	<i>B</i>
2nd year	<i>C</i>	\$3244.80	<i>D</i>
3rd year	\$3244.80	\$3374.59	<i>E</i>

a Find the value of:

i *A*

ii *B*

iii *C*

iv *D*

v *E*

b Find the total interest earned over the three years.

4 Assuming that a year has 365 days and 52 weeks, calculate the final value of the following investments:

a A \$7510 investment earns interest at 4.5% p.a. compounded annually over 6 years.

b A \$3400 investment earns interest at 3% p.a. compounded quarterly over 19 years.

c A \$9090 investment earns interest at 4.7% p.a. compounded semiannually over 11 years.

d A \$8920 investment earns interest at 3.3% p.a. compounded monthly over 5 years.

e A \$3420 investment earns interest at 4.9% p.a. compounded weekly over 9 years.

f A \$6000 investment earns interest at 5% p.a. compounded daily over 11 years.

- 5 Joan's investment of \$3000 earns interest at a rate of 3% p.a, compounded annually over 4 years. What is the value of the investment at the end of the 4 years?
- 6 Calculate the amount, A , that an investment of \$1000 compounded annually is worth after:
- a 3 years at an interest rate of 4% p.a. b 4 years at an interest rate of 9% p.a.
- 7 \$4000 is invested in a term deposit at a rate of 3% per quarter compounded monthly. Find the value of the investment after 8 years.
- 8 \$3000 is invested in a term deposit at a rate of 1% per month compounded quarterly. Find the value of the investment after 6 years.
- 9 Amelia borrows \$2400 at a rate of 6.3% p.a. compounded annually. If she pays off the loan in a lump sum at the end of 5 years, how much interest does she pay?
- 10 \$380 is invested at 2% p.a. compounded annually for 5 years. At the end of 5 years, the entire value of the investment is reinvested at 3% p.a. compounded annually for 4 more years. What is the final value of the investment at the end of the 9 years?
- 11 Kate invests \$3000 at a rate of 2% p.a. compounded annually. Find how much the investment is worth after:
- a 24 months b 18 months c 30 months
- 12 Ben borrows \$7000 at a rate of 2% p.a. compounded annually. After 2 years he makes a repayment of \$500. After another 3 years, with no further repayments, how much does Ben owe?
- 13 Xavier invests \$7000 in a term deposit with a rate of 2% p.a. compounded annually. After 3 years he withdraws \$600, and leaves the rest in the term deposit for 2 more years. How much is the investment worth after the total 5 years?
- 14 Assuming that a year has 52 weeks and 365 days, calculate how much is owed at the end of the following loan periods, given that no repayments are made:
- a \$8000 is borrowed at a rate of 5.4% p.a. compounded semi-annually for 5 years.
- b \$4000 is borrowed at a rate of 2.8% p.a. compounded quarterly for 3 years.
- c \$6000 is borrowed at a rate of 20% p.a. compounded annually for 5 years.
- d Emma borrows \$7000 is borrowed at a rate of 4.7% p.a. compounded annually for 3 years.
- e \$6000 is borrowed from a bank at a rate of 7.2% p.a. compounded weekly for 5 years.
- f \$7000 is borrowed from a bank at a rate of 3.4% p.a. compounded monthly for 6 years.
- g \$2000 is borrowed from a bank at a rate of 1.3% p.a. compounded daily for 3 years.



- 15** Lachlan borrows \$5000 at a rate of 4.5% compounded annually. After 2 years the bank increases the interest rate to 4.6%. If he pays off the loan in a lump sum at the end of 5 years, how much interest does he pay?
- 16** Sean borrows \$7000 at a rate of 5.5% p.a. compounded weekly. If she pays off the loan in a lump sum at the end of 5 years, find how much interest she pays. Assume there are 52 weeks in a year.
- 17** Assuming that in a year there are 52 weeks and 365 days, calculate the amount of interest earned on the following investments:
- a** Sally's investment of \$4200 earns interest at 2.7% p.a. compounded quarterly over 13 years.
 - b** Sally's investment of \$3070 earns interest at 4.5% p.a. compounded monthly over 7 years.
 - c** Han's investment of \$9110 earns interest at 3.2% p.a. compounded weekly over 18 years.
 - d** Luke's investment of \$6220 earns interest at 2.8% p.a. compounded daily over 11 years.
 - e** Buzz's investment of \$4920 earns interest at 5% p.a. compounded semiannually over 2 years.
 - f** Sally's investment of \$8910 earns interest at 4% p.a. compounded annually over 10 years.
- 18** Pauline borrows \$50 000 at a rate of 5.4% per annum. If she pays off the loan in a lump sum at the end of 7 years, find how much interest she pays if the interest is compounded:
- a** Daily
 - b** Monthly
 - c** Quarterly
- 19** Emma wants to invest \$1400 at 5% p.a for 5 years. She has two investment options, compounding quarterly or compounding monthly.
- a** Calculate the value of the investment if it is compounded quarterly.
 - b** Calculate the value of the investment if it is compounded monthly.
 - c** Calculate how much extra the investment is worth if it is compounded monthly rather than quarterly.

20 Maria has \$9000 to invest for 5 years and would like to know which investment plan to enter into out of the following three.

- **Plan 1:** invest at 4.51% p.a. interest, compounded monthly
- **Plan 2:** invest at 6.16% p.a. interest, compounded quarterly
- **Plan 3:** invest at 5.50% p.a. interest, compounded annually

- a Calculate the future value of the investment under Plan 1.
- b Calculate the future value of the investment under Plan 2.
- c Calculate the future value of the investment under Plan 3.
- d Which investment plan yields the highest return?

21 Mae's investment into a 20-year 2.33% p.a. corporate bond grew to \$13 600. Calculate the size of Mae's initial investment if interest was compounded:

- a Annually
- b Half-yearly
- c Quarterly
- d Monthly
- e Weekly, assuming there are 52 weeks in a year.
- f Daily, assuming there are 365 days in a year.

22 Katrina borrows \$6500 at a rate of 6.6% p.a. compounded semi-annually. If she pays off the loan in a lump sum at the end of 5 years, find how much interest she pays.

23 Frank is working out the compound interest accumulated on his loan. He writes down the following working:

$$A = 6000 \left(1 + \frac{0.08}{4} \right)^{(7 \times 4)}$$

- a How much did he borrow in dollars?
- b What is the annual interest rate as a percentage?
- c Is the interest being compounded weekly, monthly, quarterly or annually?
- d For how many years is he accumulating interest?
- e How much interest does he pay?



Spreadsheets

24 The following spreadsheet shows the balance in a savings account in 2011, where interest is compounded quarterly:

	A	B	C	D
1	Quarter	Balance at the beginning of quarter	Interest	Balance at the end of quarter
2	1	\$2000	\$20	\$2020
3	2	\$2020	\$20.20	\$2040.20
4	3	\$2040.20	\$20.40	\$2060.60
5	4			

- a Calculate the quarterly interest rate.
- b Complete the table for quarter 4.
- c Use a spreadsheet to find how many quarters after the beginning of 2011 the balance will be double the initial investment of \$2000.

25 The following spreadsheet shows the balance in a savings account in 2011, where interest is compounded monthly:

	A	B	C	D
1	Month	Balance at the beginning of month	Interest	Balance at the end of month
2	January	\$1000	\$20	\$1020
3	February	\$1020	\$20.40	\$1040.40
4	March	\$1040.40	\$20.81	\$1061.21
5	April	\$1061.21	\$21.22	\$1082.43
6	May			

- a Use the numbers for January to calculate the monthly interest rate.
- b Complete the table for the month of May.
- c Use a spreadsheet to calculate the total amount of interest earned over the year.

26 The following spreadsheet shows the balance in a savings account where interest is compounded quarterly:



	A	B	C	D
1	Quarter	Balance at the beginning of quarter	Interest	Balance at the end of quarter
2	1		\$100	\$5100
3	2	\$5100		\$5202.00
4	3	\$5202.00	\$104.04	
5	4	\$5306.04	\$106.12	\$5412.16

- a Calculate the quarterly interest rate, correct to two decimal places.
- b Complete the table.

27 The following spreadsheet shows the balance in a savings account in 2013, where interest is compounded monthly:

	A	B	C	D
1	Month	Balance at the beginning of month	Interest	Balance at the end of month
2	July	\$3000	\$30	\$X
3	August	\$3030	\$30.30	\$3060.30
4	September	\$3060.30	\$Y	\$3090.90
5	October	\$Z	\$30.91	\$3121.81
6	November	\$3121.21	\$31.22	\$3153.03

Complete the table.